Practice workbook · Interpret a menstrual-cycle graph of hormones or lining thickness

Graphs of hormone concentration or uterine lining width across a 28-day cycle.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-MENSTRUAL-CYCLE-GRAPH

1. Question 4 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

4 (a) State three methods of birth control.

(3)

1 .....

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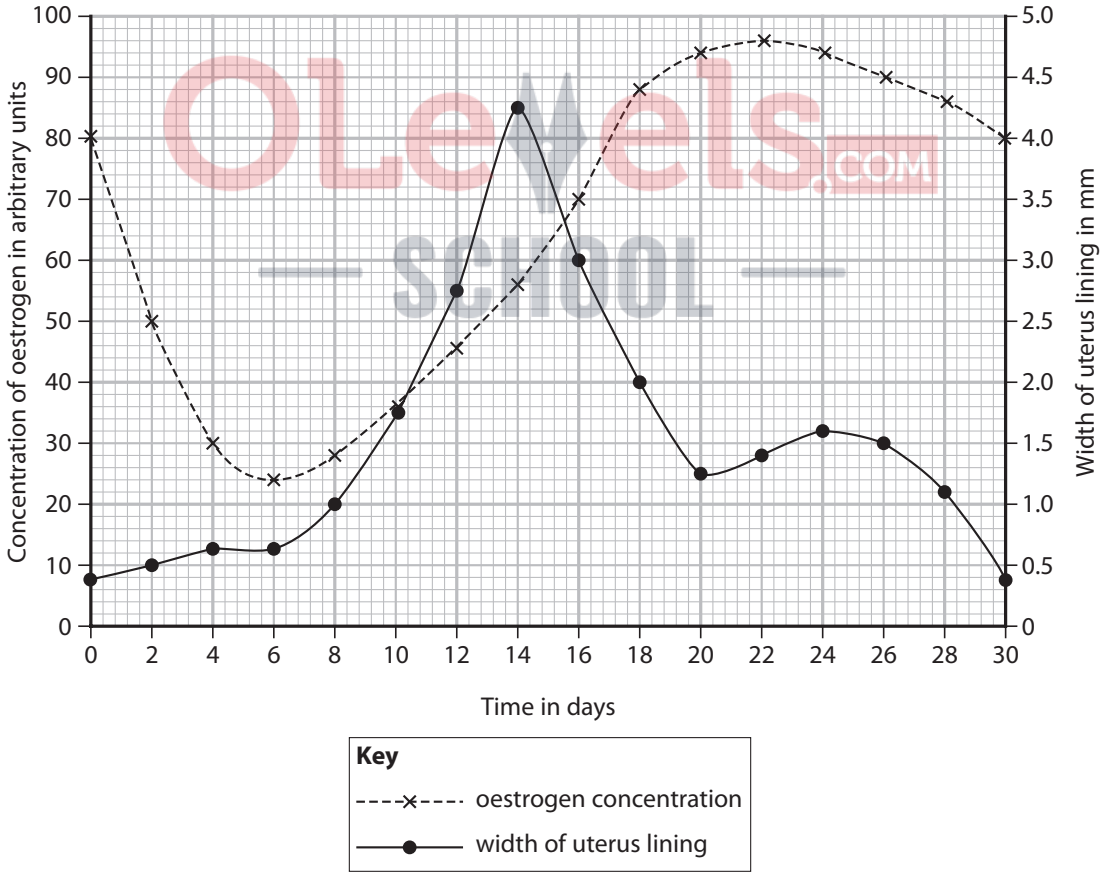
2 .....

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(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.



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1. Question 4 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

(2)

change in width = ... mm

(ii) Explain the importance of this change in width.

(3)

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(iii) Determine during which days menstruation occurred.

(1)

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

(3)

(Total for Question 4 = 12 marks)

1. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

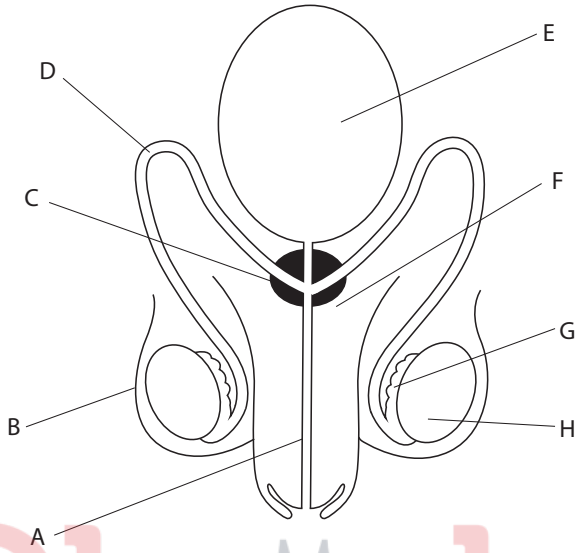
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2. Question 7 · part b.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.i only.

7 (a) The diagram shows the male reproductive system.



The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

(4)

Description of the structure	Letter
stores sperm	
produces sperm	
produces fluid to help sperm move	
contains cells dividing by meiosis	

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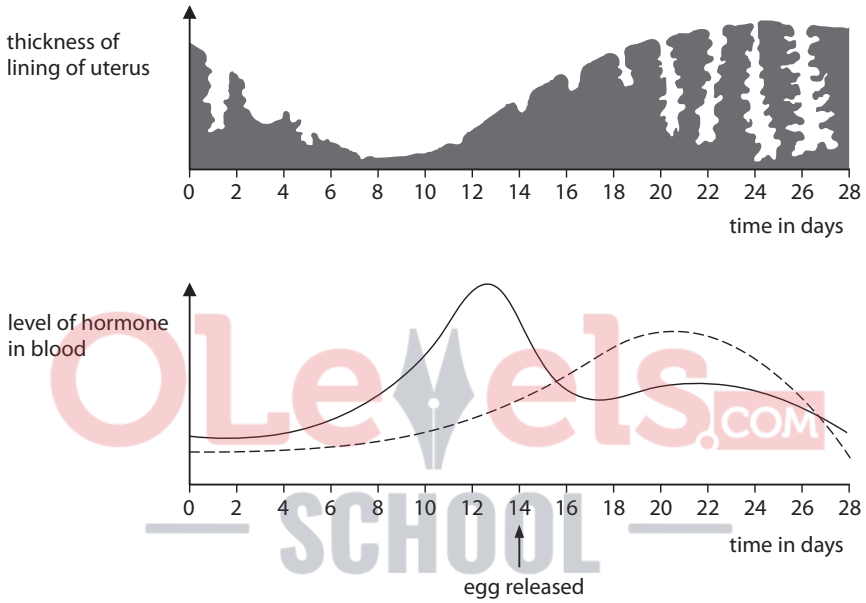
2. Question 7 · part b.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.i only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.



(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

(2)

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

(2)

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2. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

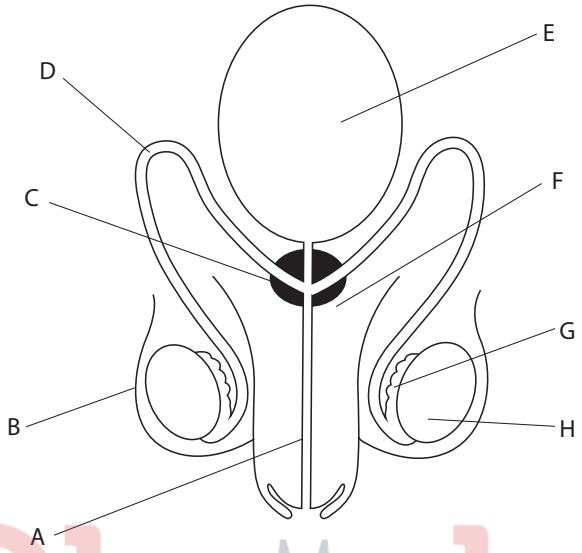
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3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

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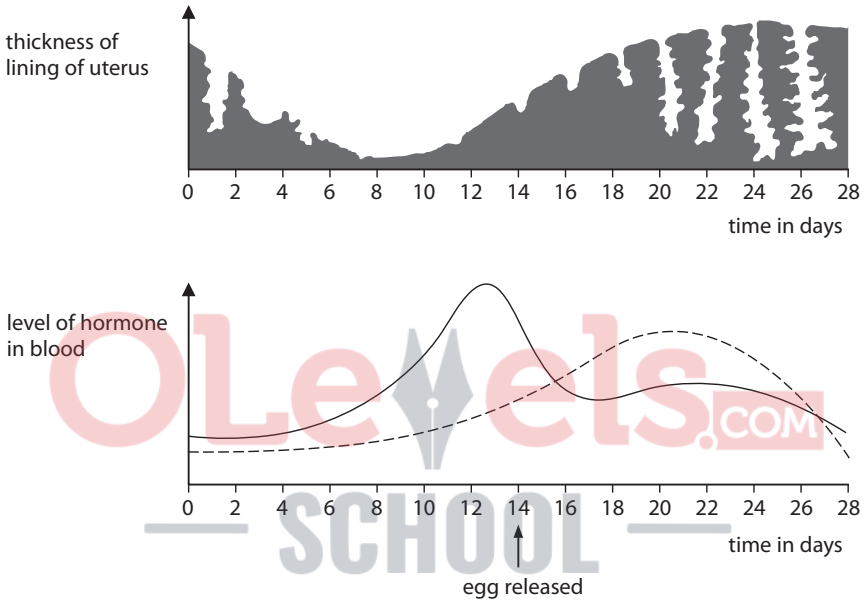
3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

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The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.



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3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(c) A synthetic oestrogen called oestradiol is used to make contraceptive pills. Natural oestrogen, which is a lipid molecule, could be used but is less effective than oestradiol when taken as a pill.

(i) Suggest why natural oestrogen is not as effective as oestradiol when taken as a pill.
(2)

(ii) Give a reason why contraceptive pills only need to be taken for 21 days out of the 28-day cycle.
(1)

(Total for Question 7 = 11 marks)

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Working and answer

4. Question 4 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

4 (a) State three methods of birth control.

(3)

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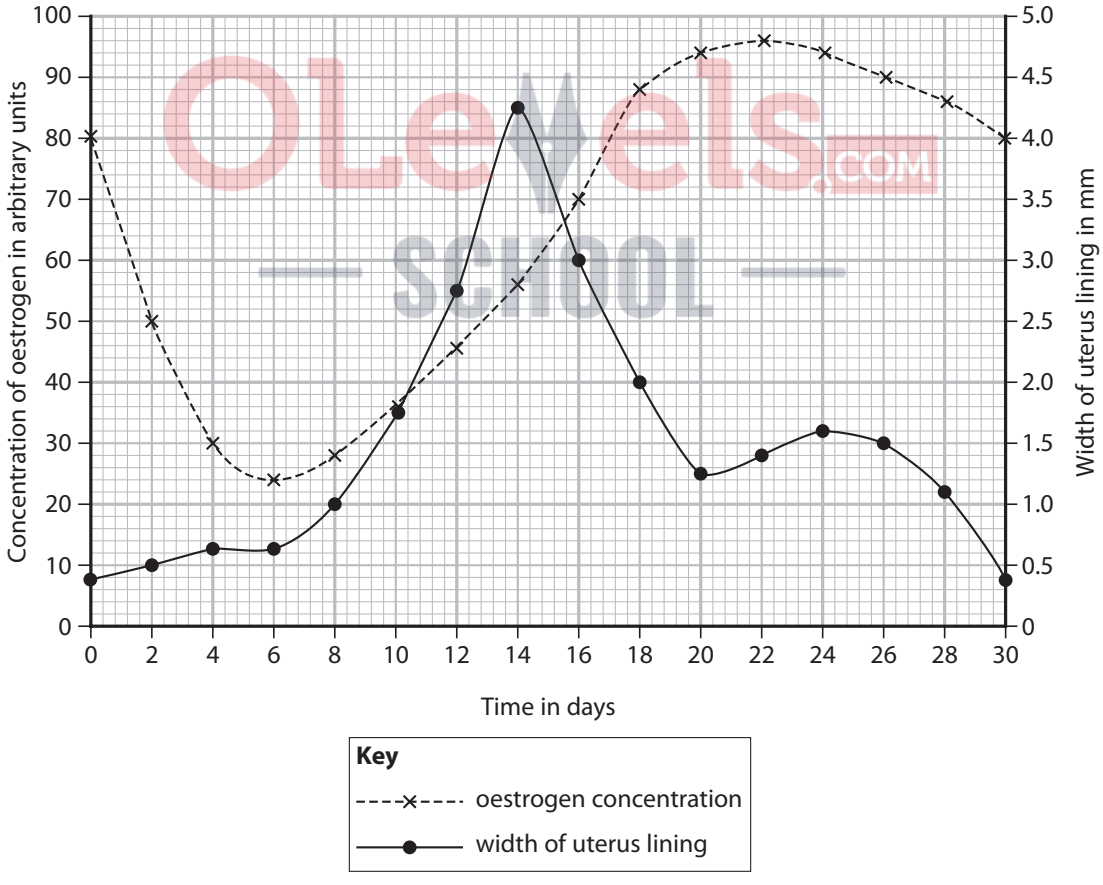
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(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.



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4. Question 4 · part b.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

(2)

change in width = ... mm

(ii) Explain the importance of this change in width.

(3)

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(iii) Determine during which days menstruation occurred.

(1)

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

(3)

(Total for Question 4 = 12 marks)

4. Question 4 · part b.iii · 1 mark · Working and answer

Working and answer

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5. Question 4 · part b.iv · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.iv only.

4 (a) State three methods of birth control.

(3)

1 .....

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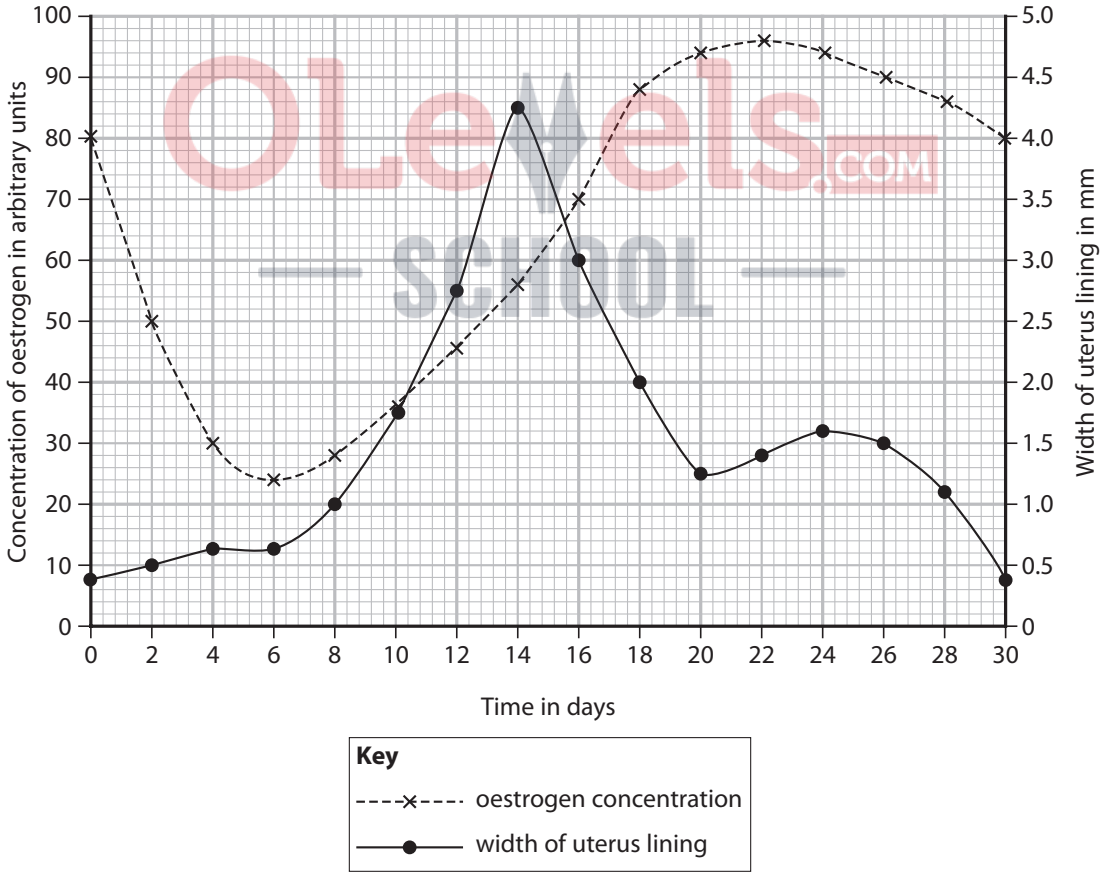
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5. Question 4 · part b.iv · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.iv only.

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(2)

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(1)

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(3)

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(Total for Question 4 = 12 marks)

5. Question 4 · part b.iv · 3 marks · Working and answer

Working and answer

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